

Symetra: Visual Analytics for Tuning Symbolic Execution Engines

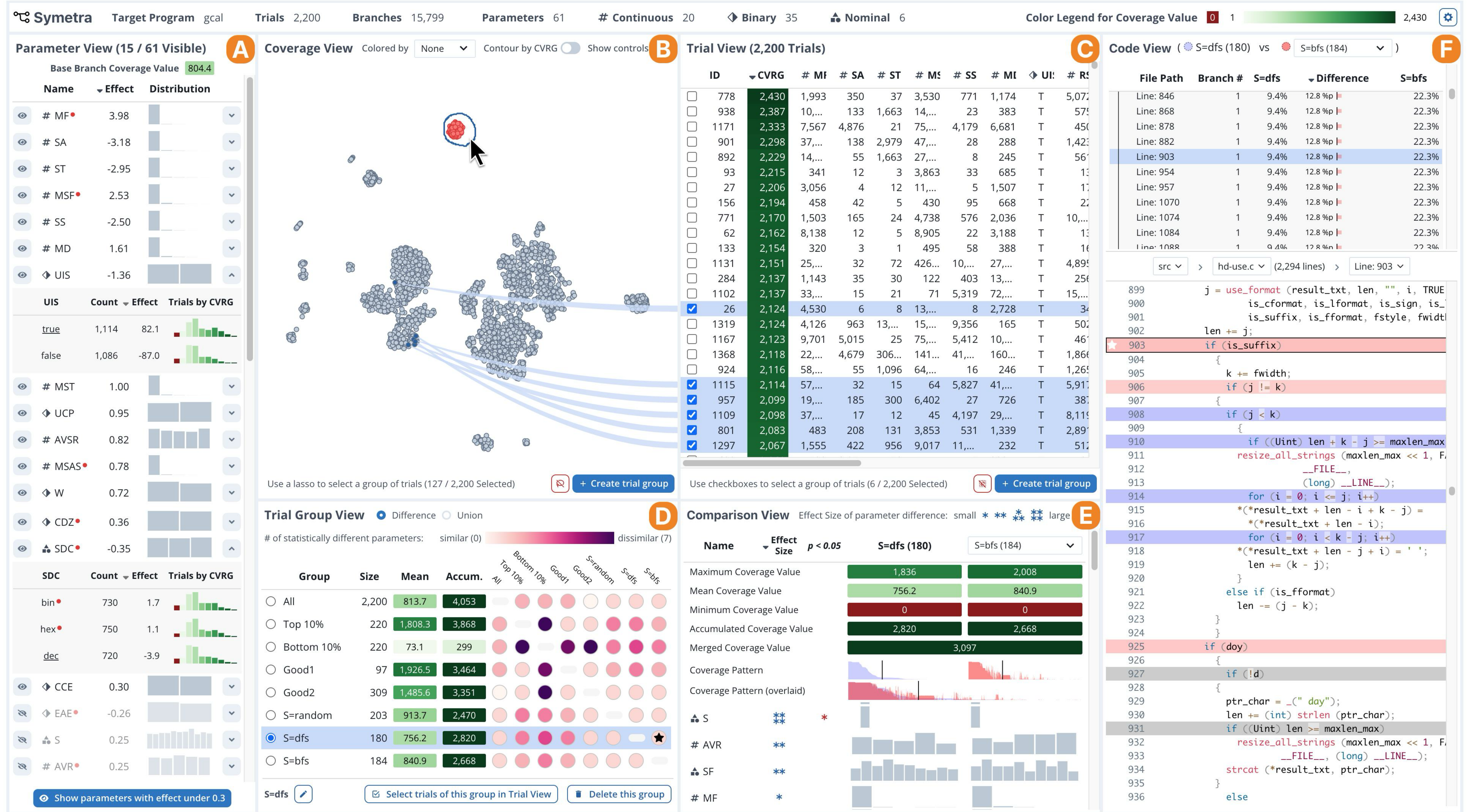
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Interface Design



- (A) **Parameter View** ranks parameters by impact on branch coverage
- (B) **Coverage View** visualizes branch coverage patterns using UMAP
- (C) **Trial View** displays individual trial configurations and coverage values
- (D) **Trial Group View** manages groups for collective analysis
- (E) **Comparison View** compares groups by coverage and parameters
- (F) **Code View** shows coverage differences in source code

Summary

Complex Parameter Tuning Challenge in Symbolic Execution

- Tens of parameters
- Complex, non-linear relationships between parameters and coverage

Symetra: First Visual Analytics System for Parameter Tuning

- Understand parameter impact
- Analyze coverage patterns
- Compare configurations systematically

Use Case

Setup

- Target program: gawk with 10,720 total branches
- Testing 800 trials per experiment

Iterative Refinement Procees

- Iteration 1: Filter out Parameters with Little Effects
- Iteration 2: Remove Parameter Values Commonly Used in Failed Trials
- Iteration 3: Redefine Defaults

Results

Stage	Branch Coverage (Total Trials)	The number of Failed Trials	Branch Coverage (First 30 Trials)
Initial	3,832	128	3,411
Iteration 1	4,093 (6.8%↑)	22 (82.8%↓)	3,651(7.0%↑)
Iteration 2	4,379 (7.0%↑)	0 (100%↓)	3,704 (1.5%↑)
Iteration 3	4,502 (2.8%↑)	0	3,813 (2.9%↑)

Limitations & Future work

Limitations

- A post-hoc analysis tool without real-time engine integration
- Single-experiment comparisons rather than cross-program analysis

Future work

- Real-time integration with symbolic execution engines
- Multi-level comparative analysis across programs and tuners

